

Biotechnology Internship — 2026



Recent Advancements in Genetic Engineering: CRISPR-Based Genome Editing and Gene Therapy

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Recent Advancements in Genetic Engineering:

CRISPR-Based Genome Editing and Gene Therapy

1. Introduction

Biotechnology may be characterized as the deliberate utilization of living organisms, cells, biological materials, or chemical agents in order to generate products and technologies that yield benefits for society as well as for the planet as a whole. The range of domains in which biotechnology finds application is remarkably extensive, encompassing areas that stretch from healthcare provision and food production all the way through to ecological management, industrial processes, and scientific research purposes. Among the areas of biotechnology that hold the greatest promise, genetic engineering stands out as particularly significant. This constitutes a multifaceted procedure that entails modifying the genotype of a particular organism so as to produce a desired protein or to bring about specific alterations (Buchan & Skelton, 2020). Its significance to biotechnology has grown considerably, given that numerous organisms possess characteristics that can be attained solely by means of interventions carried out at the level of DNA. In earlier periods, genetic engineering depended chiefly upon recombinant DNA techniques, which required the isolation of fragments through the action of restriction enzymes and their subsequent insertion into a different DNA molecule (Wiedemann, 2012). Although these approaches demonstrated considerable efficiency, they nonetheless suffered from constraints with respect to their ability to target particular DNA sites. The emergence of novel programmable genome-editing instruments brought about a fundamental transformation in the manner in which researchers engage with genomes. Such tools enable investigators to introduce precise modifications into DNA and RNA, thereby resulting in the alteration of particular proteins and, in the final analysis, the modification of living organisms (Villiger et al., 2024). The CRISPR-Cas technology was originally identified in the context of the prokaryotic immune system. By employing RNA to direct the enzyme toward specific sequences, scientists were able to adapt it into a programmable enzyme capable of cleaving double-strand DNA (Jinek et al., 2012). Owing to this programmable characteristic, the CRISPR-Cas9 method constitutes a potent instrument whose applications reach well beyond conventional double-strand DNA-breaking (Jinek et al., 2012). From 2012 onward, the discipline of genome engineering has undergone a truly remarkable process of evolution. Investigators have moved past conventional DNA targeting and editing toward more advanced forms of modification, including base editing and prime editing, RNA targeting and editing, as well as epigenome editing (Villiger et al., 2024). In addition, recent years have witnessed notable advances in the technologies employed for delivery. The principal objective underlying these innovations has been to effect a transition in genetic engineering, moving it away from the simple cutting of DNA and toward the targeted and programmable correction of mutations associated with disease.

2. Traditional Genetic Engineering vs. Modern Genome Editing

Although traditional genetic engineering and modern genome editing share the ultimate goal of making changes in biological organisms, these two methods differ in terms of ease and

precision of making these changes. In particular, the process of making recombinant DNA molecules requires isolating the gene of interest and cutting this gene with special enzymes called restriction endonucleases and DNA ligase. After the gene is isolated, it is inserted into another DNA molecule with the help of vectors and transferred into a certain cell. This technique has proved to be highly effective and has enabled the production of various recombinant proteins, including insulin, vaccines, and many other useful substances. However, gene insertion, as a part of the process of making recombinant DNA molecules, rarely allows making changes in a particular place in the genome. Modern genome editing makes it possible to make changes in a precise place in DNA molecules with the help of special proteins. The most well-known system allowing to edit RNA and DNA is CRISPR-Cas9. According to Jinek et al. (2012), this system uses guide RNA to direct Cas9 to a particular place in DNA, where this protein cuts DNA molecules. Thanks to this technique, scientists can knock out genes or make changes in DNA molecules if they provide cells with special RNA that will serve as a template for new DNA strands, and this process can be seen in the article of Villiger et al. (2024).

Feature	Traditional Genetic Engineering	Modern Genome Editing
Main approach	Recombinant DNA and gene insertion	Targeted sequence modification
Typical tools	Restriction enzymes, ligase, vectors	CRISPR-Cas, base editors, prime editors
Targeting	Often less precise	Programmable and site-directed
DNA change	Frequently adds an external gene	Can modify an endogenous sequence
Double-strand breaks	Not necessarily required	Cas9 commonly produces them; newer editors may avoid them
Major applications	Recombinant products, transgenics	Gene correction, research, therapeutics, crop improvement

The transition from recombinant DNA toward programmable genome editing represents a major increase in control over genetic modification.

3. CRISPR-Cas9 Technology

The CRISPR-Cas9 system traces its evolutionary roots back to an adaptive immune mechanism that is present within bacteria, a mechanism that empowers these microorganisms to recognize genetic material coming from foreign sources and to mount defensive responses against it. In their landmark 2012 study, Jinek et al. demonstrated that this system could be modified so as

to operate as an endonuclease aimed at particular DNA sequences, and in doing so they laid down the conceptual as well as the practical groundwork upon which the genome editing approaches based on CRISPR that exist today firmly rest.

The essential order of events can be illustrated in the manner that follows:

Guide RNA → recognition of the target DNA → binding of Cas9 → cleavage of the DNA → repair of the DNA by the cell → modification of the genome

The recognition of specific sequences is brought about through the action of the guide RNA (gRNA). This molecule comprises a segment that is complementary to the target DNA, and its role is to direct the Cas9 protein toward the genomic locus that is being aimed at. Because the guide sequence is capable of being modified through deliberate design, it becomes feasible to make use of a single Cas9 protein in order to target an extensive range of different genomic locations. Cas9 operates as a nuclease, which means that it is an enzyme that possesses the ability to cut DNA. In the case of the SpCas9 systems that are most commonly utilized, the recognition of the target depends in addition upon a short DNA sequence that is known as the protospacer-adjacent motif (PAM). The function of the PAM is to help Cas9 distinguish between target sites that are suitable, and it is absolutely essential if recognition is to take place in an efficient manner. Once the guide RNA in conjunction with Cas9 has succeeded in locating the target sequence, Cas9 produces a double-strand break (DSB) within the DNA. Following this, the cell must carry out the repair of this break. When considered within the context of genome editing, two principal repair pathways are of particular importance. The first among these is non-homologous end joining (NHEJ). NHEJ acts quickly to rejoin the broken ends of the DNA, but in carrying out this process it may bring about small insertions or deletions, which are generally referred to as indels. Alterations of this kind are able to disrupt the reading frame of a gene, and it is for this reason that they prove to be useful whenever the goal is gene knockout. The second of these pathways is homology-directed repair (HDR). At those times when a suitable repair template is available, the cell is capable of drawing upon the sequence information that is contained within that template in order to bring about a genetic change that possesses greater specificity. As a result, HDR is able to facilitate targeted correction or insertion, even though its efficiency tends to be lower and considerably more dependent upon the cell type and the biological conditions than is the case with NHEJ. The factors that account for the success of CRISPR-Cas9 are its programmability, its relative simplicity, and its capacity for adaptation. Nonetheless, the circumstance that conventional editing techniques based on Cas9 depend upon the generation of double-strand breaks within the DNA molecule gives rise to a variety of significant challenges, insofar as the cellular mechanisms of repair that are subsequently activated in response to such breaks are not amenable to control in a manner that is either entirely dependable or fully predictable. It was precisely these limitations and deficiencies that functioned as the principal driving force behind the development of the more recently introduced editing strategies, which are designed in such a way that they can generate precise and accurate modifications to the nucleotide sequence while simultaneously circumventing the intentional creation of the standard double-strand breaks upon which earlier methodologies rely (Villiger et al., 2024).

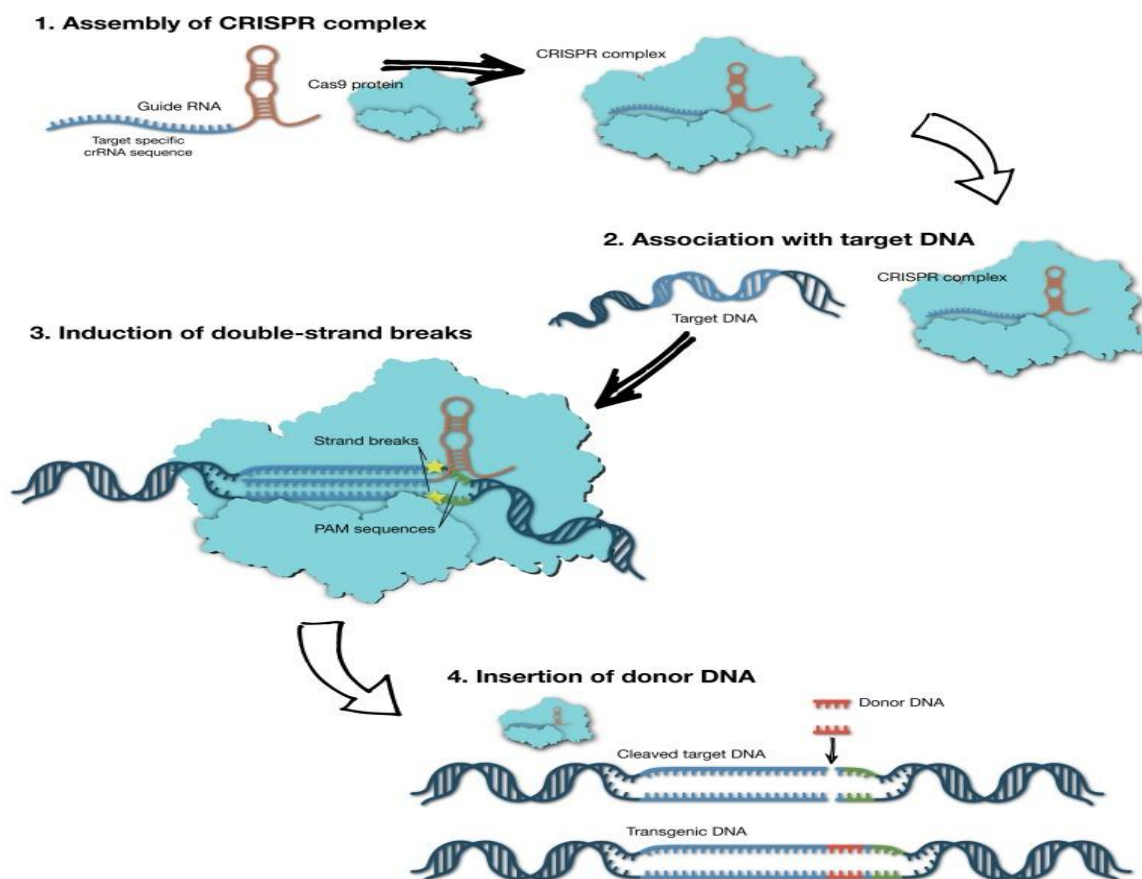


Figure 1 Mechanism of CRISPR-Cas9 genome editing showing guide RNA-directed targeting, Cas9-mediated DNA cleavage, and subsequent DNA repair.

4. Recent Advancements in CRISPR

The advancement of CRISPR technology has unfolded at an extraordinarily rapid rate, and its reach has extended considerably further than the original Cas9 nuclease platform from which the entire field first emerged. Present-day CRISPR systems are increasingly directing their attention toward refining precision, enlarging the range of targets that can be edited, guaranteeing safety, enhancing the mechanisms by which components are delivered, and widening the spectrum of therapeutic uses to which the technology may be applied. Of all these developments, base editing stands out as one of the most consequential. In place of producing a conventional double-strand break, base editors are created by joining a Cas protein that targets DNA to an enzyme that possesses the capacity to chemically alter nucleotides. Cytosine base editors are able to bring about programmable C-to-T transitions, while adenine base editors enable the conversion of A into G. This particular approach carries special importance, given that a considerable proportion of human genetic disorders are associated with single-nucleotide variants. The earliest proof-of-concept investigations demonstrated that base editing could modify nucleotides at chosen sites without requiring a double-stranded DNA break and without relying on a conventional donor DNA template (Komor et al., 2016; Gaudelli et al., 2017). More recent lines of inquiry are now extending base editing toward additional classes of mutations, even though the more recently developed transversion editors remain comparatively

less mature (2024). A second development of considerable note is prime editing. Prime editors come into being through the combination of a modified Cas9 protein and a reverse transcriptase, with a specialized prime-editing guide RNA directing the process. Rather than relying upon a standard double-strand break in conjunction with donor DNA, prime editing is capable of directly writing programmed sequence changes into a chosen region of DNA. The foundational investigation demonstrated targeted insertions, deletions, and several separate categories of point mutations within human cells, and in doing so it substantially widened the range of what could potentially be edited (Anzalone et al., 2019). In a similar manner, the discovery and engineering of further CRISPR proteins have stretched the frontiers of the discipline. Cas12 systems provide alternative means of targeting DNA and are employed both in genome editing and in molecular diagnostics (Wu et al., 2024). Cas13 systems, in contrast, are directed at RNA rather than DNA, and this opens up possibilities for transient manipulation of RNA, for the regulation of RNA, and for therapeutic approaches that centre on RNA (Zhu et al., 2024). Still another advance of significance relates to gains that have been made in specificity. Researchers have engineered altered Cas proteins, honed the design of guide RNAs, and devised more potent methods for detecting off-target activity, all in an effort to reduce unintended alterations to the genome (Wang et al., 2023). At the same time, technologies for delivery are drawing ever more sustained attention. Lipid nanoparticles are able to carry CRISPR components, among them RNA-based cargos, and they are presently being examined as nonviral delivery vehicles for in-vivo genome editing (Masarwy et al., 2024). Considered as a whole, these advances make plain that contemporary CRISPR is in the process of shifting from a single instrument for cutting genes into a wide-ranging platform that is able to carry out programmable engineering of the genome, the epigenome, and the transcriptome (Villiger et al., 2024).

5. Gene Therapy

When considered at the level of conceptual abstraction, gene therapy can be described as the intentional and deliberate delivery of genetic material into a living organism; alternatively, and from a rather different vantage point, it may be understood as the purposeful alteration of genetic material that the organism already possesses, with such interventions being carried out expressly for the aim of curing a disease or, under certain circumstances, for the aim of stopping that disease from ever arising at all. According to the specific disease that is being addressed and according to the particular therapeutic goal that is being sought, an intervention of this type may adopt any one of a number of separate strategies: substituting a gene that has been lost, providing a working copy of a gene, altering the way in which a gene is expressed, or editing a genetic sequence that has been associated with the disease under consideration. A significant conceptual difference may be established between the addition of a gene, that is to say the insertion of a gene, and the procedure that is designated as gene editing. In relation to the first of these two strategies, a functional genetic sequence is delivered into cells so that it can serve as a substitute for a gene that is either faulty or else wholly missing. Gene editing, by way of contrast, is instead oriented toward the alteration of the patient's very own genetic material. As a result of ongoing progress in CRISPR systems, this latter approach is steadily becoming increasingly feasible in practice. Therapies of a genetic nature can be sorted into categories on

the basis of the site at which the genetic alteration is performed. Where ex-vivo therapy is concerned, cells are first harvested from the patient and are subsequently modified while they remain outside of the body. Once these cells have been modified, they may undergo testing or be expanded whenever such procedures are shown to be required, and thereafter they are given back to the patient. This specific approach proves particularly beneficial when blood cells and immune cells are involved, given that such cells can be drawn out and then reintroduced. Whenever in-vivo therapy is applied, on the other hand, the therapeutic elements are delivered straight into the patient's body, and from that point they must succeed in reaching the relevant cells or tissues. A notable and groundbreaking example of an ex-vivo CRISPR therapy is exagamglogene autotemcel, which is likewise known by the names exa-cel or Casgevy. In the setting of this therapy, CRISPR-Cas9 is utilized so as to edit an enhancer region of BCL11A that resides within the patient's own hematopoietic stem and progenitor cells. The objective here is to boost the production of fetal hemoglobin, and in doing so to offset defective adult hemoglobin in sickle cell disease as well as in transfusion-dependent β -thalassemia (Locatelli et al., 2024; Frangoul et al., 2021). In the context of a phase 3 study carried out on sickle cell disease, 29 out of 30 evaluable patients managed to stay free of severe vaso-occlusive crises across a span of at least 12 consecutive months (Frangoul et al., 2024). Casgevy stands as a landmark therapy in the field of medicine, since it was the first therapy of any type to employ a CRISPR/Cas9 system and, by that means, to secure approval from the FDA. It received its initial approval in December 2023, and in July 2026 the FDA granted approval for its use in patients aged 2 years and older who are affected by sickle cell disease accompanied by recurrent vaso-occlusive crises or by transfusion-dependent β -thalassemia (U.S. Food and Drug Administration [FDA], 2023, 2026). CRISPR has convincingly shown that genome editing performed in vivo is in fact achievable. In a first-in-human trial centered on transthyretin amyloidosis, NTLA-2001 supplied CRISPR-Cas9 components through a lipid nanoparticle system and was capable of reducing serum transthyretin levels to a dose-dependent extent after the treatment had been administered (Gillmore et al., 2021).

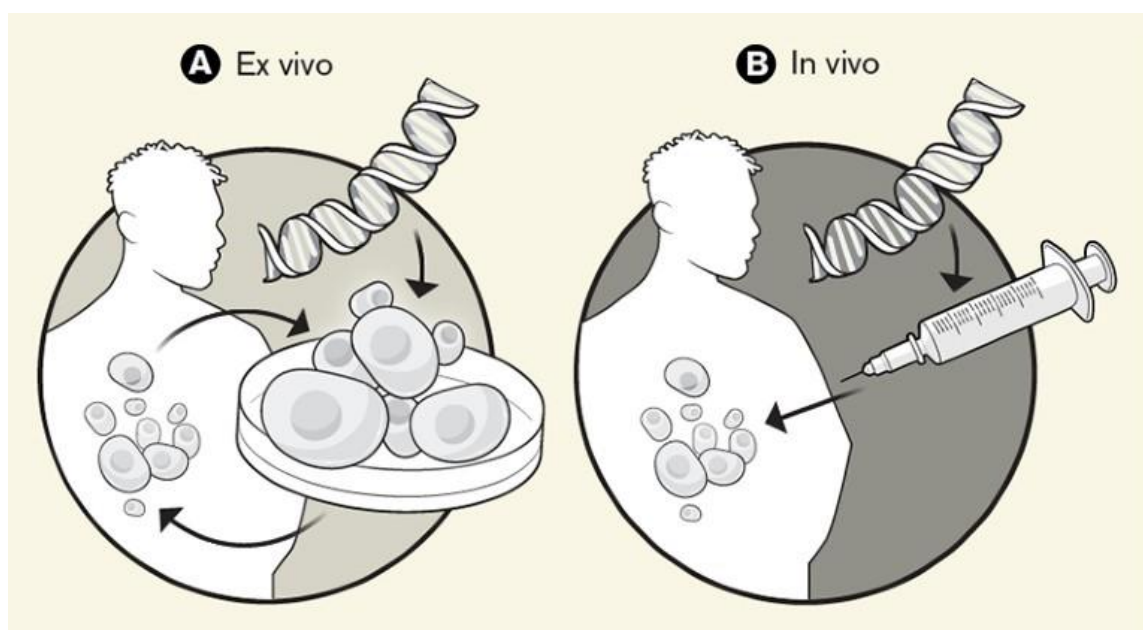


Figure 2 Basic principles of ex-vivo and in-vivo gene therapy.

6. Applications

Medicine

At the present time, the most advanced and innovative applications of CRISPR technology are to be found primarily within the field of human medicine, where they represent the forefront of current biotechnological practice. At present, investigations into genome editing are being carried out in relation to inherited blood disorders, diverse forms of cancer, metabolic diseases, infectious diseases, and a broad spectrum of additional conditions that possess a genetic basis. In the specific case of sickle cell disease, as well as β -thalassemia, the editing of hematopoietic stem cells possesses the capacity to alter the regulation of hemoglobin, and in doing so, to diminish the clinical manifestations that are associated with these diseases. Within other contexts, editing that is performed in vivo may render it possible for a gene that is responsible for causing disease to be either selectively disrupted or, alternatively, corrected. In addition, CRISPR systems are currently being examined as research instruments that serve the purpose of identifying genes linked to disease and facilitating the development of therapeutic targets.

Agriculture

CRISPR has emerged as a tool of considerable importance with regard to the improvement of crops. It is possible for researchers to modify genes that are associated with resistance to disease, tolerance of stress, nutritional quality, traits related to yield, and various other agricultural characteristics. When compared with the practices of conventional breeding, targeted editing is capable of accelerating the introduction or the study of particular genetic traits (Uranga et al., 2024). Contemporary research in the agricultural field is likewise investigating base editing, prime editing, approaches to editing that do not involve DNA, and combinations in which CRISPR is paired with other technologies, all with the aim of enhancing crop sustainability and food security.

Research

Within the realm of basic research, CRISPR makes it possible to carry out functional analysis of genes, to create models of disease, to perform genetic screening, and to investigate regulatory networks. Cas variants that regulate DNA rather than cutting it, or that instead target RNA, have served to further broaden the array of biological questions that are able to be studied.

7. Challenges and Ethical Considerations

Despite the remarkable pace at which genome editing technologies have progressed over recent years, the field nevertheless continues to be confronted with considerable challenges, both of a scientific character and of a societal character. The most prominent among these challenges is off-target editing, a phenomenon whereby CRISPR introduces modifications that were never intended, occurring at genomic locations that exhibit some degree of similarity to the target site that had originally been designated. Alterations of this sort are capable of making safety assessments substantially more difficult to carry out, and this heightened difficulty is

particularly evident when one considers the context of clinical applications. As a result, researchers are now actively devoting their efforts to developing enzymes that exhibit greater fidelity, to refining the strategies by which guides are designed, and to establishing comprehensive detection methodologies that function across the entirety of the genome (Wang et al., 2023). A second obstacle that demands attention is the matter of delivery. It is of fundamental importance that the components necessary for editing be delivered to the appropriate cells in sufficient quantities, and that this delivery be achieved without producing levels of toxicity that would be regarded as unacceptable. Viral vectors, lipid nanoparticles, approaches to delivery based on RNA, and ribonucleoprotein systems each offer their own particular advantages; at the same time, however, every one of them is likewise burdened by its own distinct set of limitations. Achieving delivery that is simultaneously efficient and selective with regard to tissues located beyond the liver continues to represent a challenge of especially formidable proportions (Masarwy et al., 2024). Further issues warranting attention include immune responses, the intricate complexities that are inherent in manufacturing, the substantial expense associated with treatment, and the disparities that exist in terms of access to sophisticated genetic medicines. The costs attached to cell and gene therapies are capable of erecting formidable barriers to accessibility on a worldwide scale, which in turn raises pressing questions concerning the equitable distribution of the benefits that such technology confers. When viewed from an ethical perspective, genome editing that influences the germline and can therefore be inherited calls for particularly careful deliberation, given that the alterations introduced into embryos or into reproductive cells may be transmitted to generations that have yet to come into existence. The World Health Organization has emphasized the necessity of putting in place robust international governance, mechanisms for oversight, transparency, registries, and ethical safeguards in connection with human genome editing (WHO, 2021). When these considerations are taken together, it becomes evident that scientific capacity must be accompanied by regulation that is responsible in nature and by meaningful engagement with the public.

8. Future Perspectives

It is probable that the future trajectory of genetic engineering will be characterized by the emergence of editing technologies that are ever more precise in their operation and increasingly versatile in the range of applications they can support. The utilization of base editors alongside prime editors holds the potential to diminish the field's reliance upon conventional approaches that depend on double-strand breaks, while the discovery and development of novel Cas proteins could serve to broaden the spectrum of genomic targets as well as RNA targets that are amenable to manipulation. Furthermore, it is anticipated that ongoing advances in high-fidelity editing methodologies and in the technologies used for delivery will contribute to enhancements in both the safety profile and the therapeutic effectiveness of clinical applications. An additional direction of considerable importance is represented by the field of personalized medicine. In light of the fact that genomic sequencing is becoming progressively more accessible, it is plausible that treatments will increasingly be formulated with reference to the particular disease-causing mutation harbored by an individual patient. Methods grounded in artificial intelligence and machine learning may prove useful in a variety of tasks, including

the design of guide RNAs, the prediction of editing outcomes, the identification of off-target sites, and the selection of appropriate targets, thereby potentially rendering genome engineering a more efficient undertaking. Nevertheless, it must be acknowledged that technological advancement on its own will not suffice to guarantee that clinical translation will be achieved successfully. A number of considerations will continue to be indispensable, among them the long-term monitoring of safety, the exercise of regulatory oversight, the question of affordability, the availability of sufficient manufacturing capacity, and the assurance of equitable access. It is conceivable that the convergence of precise editors, enhanced delivery systems, computational tools, and responsible governance frameworks may ultimately enable genetic engineering to transition away from its status as an experimental form of treatment and toward a substantially broader role in clinical practice..

9. Conclusion

Genetic engineering has come a long way since wide applicability recombinant-DNA methods. More programmable and specific techniques for altering particular DNA parts have been developed, with CRISPR-Cas9 being a huge milestone in terms of easy and versatile targeted genome editing. Following the initial breakthrough, base editing, prime editing, and new Cas proteins such as Cas12, Cas13, and others, as well as various delivery vectors, have been discovered or engineered, enabling an even broader range of alterations and tweaks to be made to DNA sequences. There is one major sign that the field has transitioned from interesting but still essentially academic research to a practical tool for medical applications, which is the translation of CRISPR into a therapy. Specifically, Casgevy's success means that gene editing aberrations in the genome of blood cells resulting in devastating blood diseases may be effectively regulated and corrected in a controlled manner. In addition, numerous studies have demonstrated the possibility of making such corrections within the body. Nevertheless, there are still considerable challenges to be addressed before gene editing may be utilized in a broader sense, such as off-target effects, delivery mechanisms, financial accessibility, ethical concerns, and regulatory issues. To conclude, the field of genetic engineering is currently at the edge of a new era of precision medicine, yet further research and discoveries are required.

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